

Table 1: Table 2. Proposed Ecological Hypotheses and Path Expectations

Tested across baseline SEM configurations (DAG1A–1C) and the two-stage interaction model (DAG1D)

Code	Ecological Mechanism	Target Structural Path	Expected Sign	Target DAGs	Supporting Literature
H1: Bottom-up					
H1a	Early marine prey availability promotes juvenile growth	PreyNCC -> Growth	+	DAG1A, DAG1B, DAG1C	Crozier et al. (2021)
H1b	Early marine growth enhances early ocean abundance (CPUE)	Growth -> Abundance	+	DAG1A	Wells et al. (2020)
H1c	Early marine growth directly increases smolt-to-adult survival	Growth -> SAR	+	DAG1B, DAG1C	Crozier et al. (2021)
H1d	Early marine abundance dictates adult return strength	Abundance -> SAR	+	DAG1A, DAG1B, DAG1C	Wells et al. (2020)
H1e	Subarctic marine productivity enhances late marine survival	PreyAK -> SAR	+	DAG1A, DAG1B, DAG1C	Wells et al. (2023)
H2: Top-down					
H2a	Early ocean predator pressure reduces juvenile abundance	PredNCC -> Abundance	-	DAG1A, DAG1B, DAG1C	Wells et al. (2025)
H2b	Early ocean predation imposes delayed mortality on return rates	PredNCC -> SAR	-	DAG1C	Wells et al. (2025)
H2c	Late marine predation acts as a bottleneck on adult survival	PredAK -> SAR	-	DAG1A, DAG1B, DAG1C	Crozier et al. (2025)
H3: Moderated					
H3a	Alternate prey buffer early juvenile salmon from NCC predators	PredNCC * AltPreyNCC -> Abundance	+	DAG1D (Two-Stage)	Wells et al. (2023)
H3b	Buffer prey induce functional switching by sea lions/sharks	PredAK * AltPreyAK -> SAR	+	DAG1D (Two-Stage)	Wells et al. (2023)
H3c	Thermal conditions alter spatial overlap and predation intensity	Pred * SST -> Survival	+/-	DAG1D (Two-Stage)	Wells et al. (2025)